

Exam III

Review Topics

Section	Topics	Problems
10.1	Sequences Convergence Series Definition Convergence Geometric Series Definition Sum (Formula)	Page 365 6, 15
10.2	Tests for Convergence a. Divergence Test b. Integral Test (s.n.n.t.) c. Comparison Test (s.n.n.t.) d. Ratio Test (s.n.n.t.) e. n-th Root Test (s.n.n.t.)	Page 371 24, 29, 43, 47, 49, 52
10.3	MacLaurin Series	Using the definition, find the first 3 non-zero terms of the MacLaurin series of a. e^{2x} b. $\tan x$ c. $\frac{x}{1+x}$
10.4	Operation with Series Composition Products Term-by-Term Differentiation Term-by-Term Integration	Page 380 6, 8, 9, 14
10.5	Applications Alternating Series Approximation and Error Estimates Taylor Series Approximation and Error Estimates	Page 386 1, 8, 13, 17 Using a series expansion, find the value of $f(a)$, accurate to four decimal places a. $f(x) = \sin x$ $a = 0.2$ b. $f(x) = e^x$ $a = -0.1$
10.6	Fourier Series Definition Calculation of Fourier Coefficients	Page 397 4
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